Lesson 6: Victory Condition

TIC-80

VIDEO GAME CREATION

This week: a real win condition. Your job is to track how many enemies are left and show "You Win!" the moment the last one is destroyed - completing your game's Level 1 MVP milestone.

Open your own [YourName]Game project from Lesson 5 - don't start a new one. Everything this week builds on the row of enemies you already have.

Tip: no new assets this week - "You Win!" is shown with each engine's own basic text tools, not an image.

Setup

1. Launch TIC-80 and load your cart from Lesson 5.
2. Press Esc to reach the code editor.

Level 1 - Minimal (MVP)

The moment the last enemy is destroyed, "You Win!" appears on screen. This completes your game's Level 1 MVP milestone - a complete, playable core loop.

3. Add enemies_remaining=5 near the top of your code, alongside your enemies table.
4. Inside the bullets loop's hit branch (where an enemy's alive is set to false), decrement enemies_remaining.
5. At the bottom of TIC(), add an if check that calls print("You Win!", ...) when enemies_remaining reaches 0 (code below).
6. Press Esc, type run, press Enter, then shoot every enemy to test - "You Win!" should appear the instant the last one is destroyed.

```
enemies={}
for enemy_count=0,4 do
 table.insert(enemies,{ex=20+enemy_count*20,ey=10,alive=true})
end
enemies_remaining=5
bullets={}

function TIC()
 cls(0)
 if btn(3) then x=x+1 end
 if btnp(4) then
  table.insert(bullets,{bx=x,by=y})
```

```
 end
 for i=#bullets,1,-1 do
  local b=bullets[i]
  b.by=b.by-2
  local hit=false
  for j=1,#enemies do
   local e=enemies[j]
   if e.alive and b.bx<e.ex+8 and b.bx+8>e.ex and b.by<e.ey+8 and
b.by+8>e.ey then
    e.alive=false
    enemies_remaining=enemies_remaining-1
    hit=true
   end
  end
  if hit then
   table.remove(bullets,i)
  elseif b.by<0 then
   table.remove(bullets,i)
  else
   spr(2,b.bx,b.by,0)
  end
 end
 for j=1,#enemies do
  local e=enemies[j]
  if e.alive then spr(3,e.ex,e.ey,0) end
 end
 spr(1,x,y,0)
 if enemies_remaining<=0 then
  print("You Win!",84,60,12)
 end
end
```

Level 2 - Complete (Alpha/Beta)

The win text appears exactly once, exactly when the last enemy is destroyed - not early, not late, and not repeatedly.

- Destroy all 5 enemies and confirm "You Win!" appears exactly when enemies_remaining reaches 0, staying on screen from then on.
- This already works correctly because enemies_remaining is only ever decremented inside the same guarded branch (e.alive and ...) that already prevents a single enemy from being counted as destroyed more than once.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: import the provided sfx_win.wav into a new SFX slot (TIC-80's Surf/import tools accept an external .wav) rather than designing a new effect from scratch, then call sfx(slot_number) the moment enemies_remaining first reaches 0 - guard it with a has_won flag so it doesn't replay every single frame while the win text is showing.
- Visual: use print()'s extra parameters (scale, fixed width) or draw a rectangle behind the text with rect() first, so the message stands out more.
- Structure: keep enemies_remaining declared once at the top with your other game state, and only ever change it from inside the collision check - the same principle you've already been applying to enemies_alive and bullets. (TIC-80 has no object system at all,

so there's no separate "GameManager" object to introduce here - a top-level global variable already is this engine's native equivalent.)

If Something's Wrong

- "You Win!" never appears: confirm enemies_remaining is actually being decremented inside the hit branch, not just enemies[j].alive.
- "You Win!" appears immediately: confirm enemies_remaining starts at 5 (or your actual enemy count), not 0.
- Text flashes then disappears: confirm the print() call is guarded by enemies_remaining<=0, which stays true forever once reached - if it disappears, something else may be resetting the variable.

Want More?

Student Resource List > TIC-80 section, or tic80.com